

What's new in the Microsoft Game Development Kit (GDK)

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The Microsoft Game Development Kit (GDK) is a complete set of tools, APIs, extensions, and programming models that can be used across current and future Microsoft Gaming platforms and initiatives. The GDK's design and features are driven by your feedback about how you like to develop games. For a complete introduction, see [Welcome to the Microsoft Game Development Kit](#).

In addition to many bug fixes, this release includes the following new features.

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Developer tools

Visual Studio 2026 GDK support

Beginning with the April 2026 GDK, Visual Studio 2026 is now supported for GDK development. Both the Professional edition and the Enterprise edition are supported.

Visual Studio project templates are available as follows.

- Console development templates are installed by the GDK.
- PC development templates can be installed from [Microsoft GDK Templates](#).

Visual Studio 2026 introduces a new servicing and support model based on yearly releases and a two-year servicing timeline. For more information, see [Visual Studio Product Lifecycle and Servicing](#).

For more information about Visual Studio 2026, see [Visual Studio 2026 is here: faster, smarter, and a hit with early adopters](#).

MSIXVC2 packaging format preview

MSIXVC2 is a new packaging format for PC games built with the GDK. It delivers significantly smaller base game packages and content updates, faster packaging, and a simplified upload workflow as compared to the original MSIXVC format. MSIXVC2 is available as a preview starting with the April 2026 GDK.

Key improvements include the following:

- **Smaller content updates.** Byte-level change tracking eliminates alignment padding and chunk-boundary constraints, reducing update sizes by 64–94% compared to MSIXVC.
- **Faster packaging.** makepkg2 pack delivers 2x–8x speedups, depending on title size.

- **Reduced package overhead.** Over 90% reduction in fixed hash-tree overhead; supports partial download instead of full re-download with every update.
- **Built-in compression.** Transparently compresses assets for transport and storage; installed uncompressed on the player's device.
- **Per-version encryption.** Each package version is encrypted with its own keys, keeping content secure until release.
- **One-command upload.** Combines packaging and uploading into a single step with `makepkg2 upload`.

For more information, see [MSIXVC2 packaging overview](#).

NOTE: We recommend not using this new packaging format in production. MSIXVC2 submissions to the public will be blocked by certification tests until the general availability (GA) release, which is currently targeted for the October 2026 GDK.

Submission Validator update

Submission Validator version 10.0.26100.7798 is now available. Download the latest version from <https://aka.ms/currentsubvalzip>.

IMPORTANT: Starting with this release, the legacy download location at aka.ms/gdmdl has been retired. Submission Validator is now available exclusively from <https://aka.ms/currentsubvalzip>. The previous version (10.0.26100.6877) remains valid until April 30, 2026.

NOTE: As announced in March 2025, updating Submission Validator is now simplified. Programmatic, unauthenticated updates are available directly in your pipeline. For download and update instructions, see [Updating Submission Validator](#).

This release includes the following changes.

- **MSIXVC2 developer preview packages now report a FAILURE.** Submitting MSIXVC2 packages to certification isn't supported during the preview period. This aligns with the certification test block that's described in the [MSIXVC2 packaging format preview](#) section above.
- **ILI_ArchitectureArm64Mismatch** no longer reports a FAILURE for Arm64EC (emulation compatible) binaries.
- The **ILI_ArchitectureAmd64Mismatch** and **ILI_ArchitectureX86Mismatch** warnings now only apply to executables that are listed in the `MicrosoftGame.config` file.
- If Submission Validator encounters a crash during symbol bundling, it now reports an **ILI_SymbolHandlingInitError** INFO message and continues validation instead of exiting early.

XblPcSandbox.exe command-line tool improvements

The **XblPcSandbox.exe** command-line tool now has quality-of-life improvements to enhance the sandbox-switching experience for PC game developers.

Key improvements include the following:

- **Clearer sandbox switch output.** **XblPcSandbox.exe** now displays the current sandbox value before and after a switch operation. This display makes it immediately obvious if the intended sandbox was applied. In addition, an indeterminate progress spinner is now available to show when the tool is actively working.

- **New `/retail` switch.** A dedicated `/retail` command-line option is now available to make it easier to switch your PC back to RETAIL mode. Previously, this was only documented as passing RETAIL as the sandbox name, which could be missed.
- **Automatic sandbox casing correction.** The sandbox switching logic now automatically corrects the casing of sandbox names (for example, retail to RETAIL). This logic reduces errors that are caused by case sensitivity. Most sandbox names are in all capital letters.
- **New `/feedback` switch.** A `/feedback` option is now available that deep links directly to the Feedback Hub, making it easier to report issues or provide feedback about the sandbox-switching experience.
- **Enhanced Help text with troubleshooting guidance.** The Help usage text now includes troubleshooting suggestions and a link to the troubleshooting documentation. Solutions to common sandbox issues are now available directly from the command line.

For more information, see [PC Sandbox Switcher \(XblPCSandbox.exe\)](#).

Xbox PIX support for file mapping in DirectStorage events

Xbox PIX now supports packaged file mapping in DirectStorage events. This matches the existing File IO event mapping behavior and properly indexes the list of files along with their offsets and descriptions.

Flexible MicrosoftGame.config Editor

MicrosoftGame.config Editor no longer enforces game configuration schema validation when opening files. If a file is valid XML, the editor attempts to load it by using the correct node structure, even if the configuration itself is invalid. This allows invalid configuration files to be opened and fixed in the editor instead of failing to load.

System

Native ARM64 build GDK support preview

The GDK now supports a preview of native ARM64 build libraries for binaries that ship with the GDK. These binaries include the [PlayFab Unified SDK \(v2\)](#), the Xbox Services API, the Xbox Authentication Library, Game Chat 2, and [xCur1](#). This support enables you to produce fully native ARM64 game builds.

For more information see, [A simplified GDK directory structure](#).

NOTE: This preview is intended for testing only. We recommend not using it in production.

Samples

For a complete list of samples included with the Microsoft Game Development Kit (GDK), see [Microsoft Game Development Kit samples](#).
